

Lab 5 Worksheet — Protocols and Ports — Building the A+ Port Reference

Name: ____ Date: ____

Exam objective: Identify the TCP/UDP ports, transport protocol and secure replacement for every protocol on the CompTIA A+ Core 1 objective 2.1 list.

Record as you go

Step	What you did	What you observed
1	Open https://killercoda.com/playgrounds/scenario/ubuntu in your browser and wait for the terminal prompt to appear. You have root access and nothing is installed on your own machine.	
2	Update the package list and install the networking tools you need for this lab.	
3	Look up each A+ protocol in the system services database to confirm its official port and transport.	
4	Record the port numbers for the file transfer and remote access group: FTP 20/21, SSH and SFTP 22, Telnet 23, RDP 3389.	
5	Record the mail group: SMTP 25, POP3 110, IMAP 143, and note the secure TLS variants SMTPS 587/465, POP3S 995 and IMAPS 993.	
6	Record the infrastructure group: DNS 53, DHCP 67 server and 68 client, and note that DNS uses UDP for queries and TCP for zone transfers.	
7	Prove DNS runs over UDP by querying a public resolver and observing the transport used.	
8	Confirm HTTP on 80 and HTTPS on 443 by requesting headers from a live site and reading the response code.	

Step	What you did	What you observed
9	List every listening socket on the playground with its protocol, port and owning process, and identify which are TCP and which are UDP.	
10	Start a listener on a chosen port, then scan it from the same host to see how a scanner reports an open port.	
11	Complete the reference table with a Security column marking FTP, Telnet, HTTP, SNMPv1/v2 and POP3 as insecure, each with its secure replacement named.	
12	Add the final rows for NetBIOS 137–139, SNMP 161/162, LDAP 389 and SMB/CIFS 445, then verify your table has all 16 protocols.	

Verification

Success criterion: Your table lists all 16 protocols with the correct port, transport and purpose; every insecure protocol has its secure replacement named; and ss -tulnp output confirms at least one live listening socket with its transport.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
